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ANALYTICAL MODELING OF OPTIMAL CHUNK SIZE FOR EFFICIENT TRANSMISSION IN INFORMATION-CENTRIC NETWORKING

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Abstract. *As an innovative network architecture, information-centric networking (IC-N) is potential to promote network with name-based content retrieval. In ICN, contents are divided into chunks by default. Because chunk is the basic data unit, the size of chunk dramatically affects the transmission efficiency. In this paper, an analytical model is developed to obtain the optimal value of chunk size for content transfer in ICN. Intuitively, the larger the chunk size, the better the performance of data transfer. This is because the protocol overhead of each chunk is relatively fixed. However, larger chunk usually leads to larger chunk loss rate and larger response latency. Therefore, the content transfer process is modeled by considering both the protocol overhead and network status. Three typical transmission modes are analyzed, and the expressions for the goodput are derived. By using numerical analysis, the effects of chunk size on goodput under different round-trip times (RTTs), packet loss rates and bandwidths are studied in depth. Based on the model, the recommended values of the chunk size are provided for two deployment approaches in ICN. Finally, experiments are conducted to verify the accuracy of the proposed model.*
Keywords: Chunk size, ICN, Modeling, Transmission efficiency

1. **Introduction.** Recently, most Internet traffic is governed by content retrieval applications like P2P file sharing. Users only care about the content in these applications instead of the location of the content [1]. Currently, the Internet architecture is based on a host-to-host communication paradigm. Moreover, a significant amount of traffic is generated when multiple users repeatedly download the same popular content. Therefore, a novel networking paradigm called ICN [2] has been proposed. In ICN, in-network caching is used to deal with later repeated requests, and contents are divided into chunks by default [3].

Since chunk is the basic data unit to which transmission, caching and security functions are applied [4], chunk size is crucial in transmission efficiency. On the one hand, the frequency of system operations is determined by the number of chunks. Hence, transmitting less number of chunks can reduce the system complexity and enlarge the payload ratio. However, the partial loss of a chunk may lead to the failure of the entire chunk. Moreover, the large chunk usually leads to large response latency. The larger chunk, the more inaccurate between content popularity and cached contents, and then the lower caching

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benefits [3]. On the other hand, small chunk size can decrease the chunk loss rate. However, the smaller chunk size means more number of chunks to be transmitted. It brings more computation overhead and more management overhead to the namespace. In typical ICN approaches proposed in [5-8], only after translating a chunk identifier to their locators by using name resolution service (NRS), the chunk can be routed by the locators [9]. The small chunk size leads to the over-utilization of NRS, then the deterioration of network performance.

In existing ICN studies, different recommendations have been proposed for chunk size, but only qualitative or experimental proposals are involved. The qualitative or experimental method is straightforward and easy to apply. However, this method is subjective and is hard to obtain an accurate optimal value of chunk size, which cannot guarantee to maximize transmission efficiency. Adopting the quantitative mathematical analysis method to model the content transfer in ICN can obtain a more accurate optimal chunk size. This method is more scientific and is supported by rigorous mathematical theory. To the best of our knowledge, the issue of using the mathematical analysis to find the optimal chunk size has not been investigated. Motivated by the above discussion, neither small nor large chunk size can optimize transmission efficiency. Therefore, the goal of this paper is to find the optimal chunk size for efficient transmission by mathematical model and quantitative analysis. This research involves multiple impact factors (e.g., protocol overhead, and network status), which is worthy of being studied.

The contributions of this paper are as follows.

1) A mathematical model is developed to evaluate the effects of the chunk size on content transfer in ICN. Besides, three typical transmission modes are analyzed.

2) The expressions for goodput are derived. The goodput is a function of the chunk size, bandwidth, RTT, packet loss rate and protocol overhead. Moreover, the optimal chunk size is calculated by maximizing the goodput.

3) Both theoretical and experimental results show the effectiveness of the proposed model. Based on the model, we provide the recommended values of the chunk size for two deployment approaches in ICN.

The rest of this paper is organized as follows. Section 2 reviews related work. The mathematical model is provided in Section 3. In Section 4, we present a numerical analysis of our model. Then, the experimental results are illustrated in Section 5. Finally, the conclusions are summarized in Section 6.

2. **Related Work.** With the development of ICN, the study of chunk size becomes more and more important. Chunk size greatly affects the performance of content transfer. Recently, fruitful results related to chunk size have been studied.

To mention a few, Thomas et al. [10] suggested the chunk size is the same as the maximum transfer unit (MTU) of a link. In [3], the analytical and experimental results showed that small chunk size could improve the caching performance. The authors also presented that the number of chunks should not be too large, due to the maintenance overheads and other practical considerations (e.g., hardware limitations, minimum video clip size, and protocol overhead).

CCN [11] proposed the chunk size was not necessarily so small to be fit in current MTU. An ongoing effort has already employed that the default chunk size is about 4K bytes [12]. However, this still implies a 10% overhead on the bit rate and the need to perform signature checks at a very high rate. In [13], the chunk size was the minimum data object granularity. The authors presented that small chunk sizes are preferable (e.g., to avoid paying a padding penalty for a myriad of small objects), though not necessarily so small to be fit in current MTUs. Clearly, as the chunk size determines the frequency of all system operations, the larger chunk size is also beneficial as it relaxes the system complexity. So authors considered chunk sizes of 10K bytes to be a reasonable compromise.

The authors in [14] designed and implemented a clean-slate wireless transport protocol named hop that uses reliable per-hop block transfer as a building block. Blocks could amortize many sources of overhead over a larger unit of transmission thereby could lead to an increase of overall utilization. The sources of overhead include retransmissions, timeouts and control packets. The block mentioned here is similar to chunk in ICN. Hop uses virtual retransmissions together with in-network caching to limit the overhead of retransmitting large blocks. Thus, the authors presented the default chunk size is 1M bytes.

Furthermore, Detti et al. [4] argued that reasonable chunk sizes could be in the order of hundreds of KBs (e.g., 128, 256, and 512) up to few MBs (e.g., 1, 2, and 4). It would be useful to introduce a transport protocol which is called ICTP below the application level. More and more application scenarios need precise segmentation techniques [15]. ICTP partitions chunks into smaller data units named segment. The transport protocol performs reliability and congestion control, much like TCP does for the transmission of a byte stream in current TCP/IP networks. Each segment of the chunk has a header that brings overhead. However, the reliable transmission mechanism of chunk segments avoids the failure of the entire chunk after packet loss and can support larger chunk size. In its experiment, the goodput of the 32KB case was higher than that of the 4KB case, because the chunk header overhead has a smaller impact.

According to the above researches, we can summarize it as follows. The recommended values of chunk size are few KBs (e.g., 1.5, 4, and 10) when there is no transport layer of ICN. For example, the default size of the chunk is about 4K bytes in CCN. Moreover, the recommended value can reach 1M bytes or even larger if there is a transport layer (e.g., Hop, and ICTP). However, these studies only have qualitative or experimental proposals for their scenario but lack deep mathematical analysis. Besides, these proposals rarely consider the change of network status but only give one fixed recommended value of chunk size for each scenario. However, adopting one fixed value of chunk size is unreasonable because the network status is changing even in a scenario. For example, with a specific value of chunk size, the transmission efficiency is high to transmit a chunk when the network is in good condition. Still, the same value of chunk size may lead to lousy transfer performance when the network status gets worse (e.g., when network congestion occurs). In this paper, our proposal is aiming to model the content transfer process by considering both protocol overhead and network status. Moreover, we will provide different recommended values of chunk size for different scenarios.

3. **Model Description.** We focus on the problem of finding the optimal chunk size for efficient transmission in ICN. The content transfer process is modeled to solve this problem, and goodput is taken as the evaluation indicator. Goodput is the application-level throughput. Moreover, goodput represents the number of payload per time slot, wherein the payload is transferred from a specific source to a particular destination successfully. The payload excludes protocol overhead and retransmitted data packets. In this section, we model the content transfer process and derive expressions for the goodput as functions of chunk size and other factors. The optimal chunk size is calculated for different scenarios to maximize the goodput for content retrieval.

In the last few years, the researches related to ICN have attracted considerable attention. The deployment methods used in ICN are classified into three types: overlay approach, clean-slate approach, and integration approach [16]. The clean-slate approach lacks the necessary network hardware facilities to support ICN. At this stage, ICN can only be tested by building overlays on IP or through simulation experiments [17]. Thus, our model is mainly built based on IP infrastructure but still works for future clean-slate approaches with different network status.

In this paper, we only consider the effects of the chunk size on transmission efficiency. The overhead of flow control, cache hit ratio, traffic model, caching efficiency, and security are not considered.

Firstly, basic notations are listed in Table 1.

Table 1. The meaning of notations in this paper

Notation	Meaning
O_c	Protocol overhead of a chunk
O_s	Protocol overhead of a segment
r_p	The packet loss rate of the underlying protocol
P_c	Payload size of a chunk
N_{ct}	The number of chunk transmission times until the receiver gets the entire chunk
N_p	The number of pieces the chunk is split
M	MTU of a link
R	RTT of a link
B	The bandwidth of a link

The protocol overhead of a chunk (or segment) represents the additional bits bundled with application data, e.g., the signature for security, freshness period for caching, control signaling, and structure maintenance information. Because protocol overhead is fixed for a given application and its running network, let payload size P_c denote the chunk size (the size of a chunk including overhead and payload, i.e., $O_c + P_c$) in our analysis. Besides, suppose that content is divided into fixed-size chunks, and a chunk is partitioned into fixed-size fragments or segments. In the chunk, the fragments are generated by IP fragmentation operations, and segments are partitioned by the transport layer of ICN.

Assuming r_p is fixed during the transmission of a chunk, and r_p is not related to packet size. This is because most of the modern routers queue, process and drop data at the granularity of packets. Besides, the underlying hosting protocol overhead and the loss of the request packets (such as interest packets in CCN) are much smaller than the data packets. Therefore, these factors are ignored in this paper.

In this paper, a chunk can be transmitted by means of two methods. One is shown in Figure 1(a), where the entire chunk is transmitted. The other is given in Figure 1(b), where the chunk is partitioned into a series of segments by the transport layer of ICN. Each segment is encapsulated in a datagram of the network layer.

3.1. **Chunk transmission.** In this case, the entire chunk is encapsulated in one packet (e.g., IP packet). This packet may be fragmented by the network layer (e.g., IP operations) and reassembled on the end host. We call this transmission mode CTM (chunk transmission mode) for short. The most typical example of CTM is CCN. CCN does not even have an independent transport layer. Its transport control is operated by combining supporting libraries of applications and the strategy module of the forwarding plane [18].

For a chunk with overhead O_c and payload P_c, the split pieces of each content can be expressed as follows:

$$N_p' = \left\lceil \frac{P_c + O_c}{M} \right\rceil \tag{1}$$

where N'_p denotes the number of split pieces for chunk transmission. If any of the fragments are lost, then this packet is discarded, and the entire chunk is failed to be assembled. The chunk loss rate is $r_c = 1 - (1 - r_p)^{N'_p}$. The number of chunk transmissions times N'_{ct} can be expressed as

$$N'_{ct} = \frac{1}{1 - r_c} = \frac{1}{(1 - r_p)^{N'_p}} = \frac{1}{(1 - r_p)^{\left\lceil \frac{P_c + O_c}{M} \right\rceil}} \tag{2}$$

During the delivery of the chunk for CTM, the total bits needed to be sent are

$$Bits_{CTM} = N'_{ct} * (P_c + O_c) \tag{3}$$

According to Equations (1), (2), and (3), the goodput of CTM is

$$Goodput_{CTM} = \frac{P_c}{Delay_{CTM}} = \frac{P_c}{N'_{ct} * R + \frac{Bits_{CTM}}{B}} = \frac{P_c * (1 - r_p)^{\left\lceil \frac{P_c + O_c}{M} \right\rceil}}{R + \frac{P_c + O_c}{B}} \tag{4}$$

where $Delay_{CTM}$ denotes the total time spent from sending the request to retrieving the entire chunk for CTM.

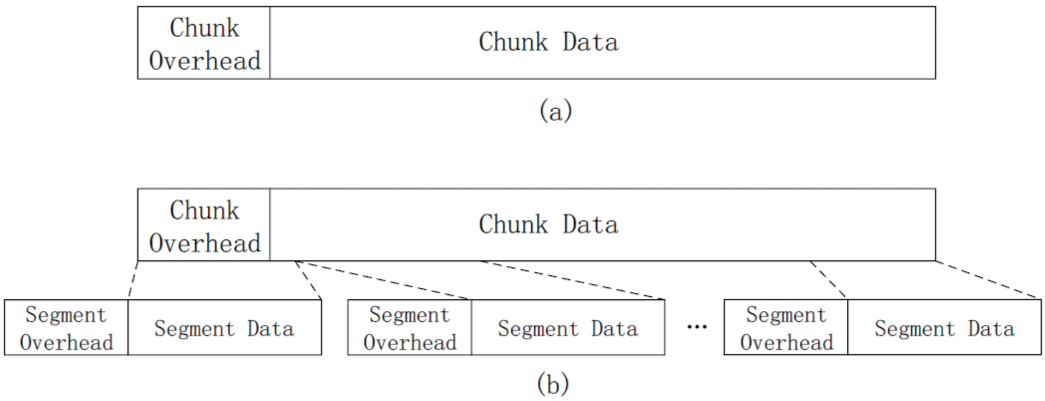


FIGURE 1. A chunk can be transmitted by means of two methods: (a) chunk transmission, and (b) segment transmission.

To figure out the optimal value of chunk size to maximize goodput, we take the derivative of goodput with respect to P_c in Equation (4). Assuming P_c is the unique variable and other factors, such as O_c, r_p, M, R, B, are constants. The rounding of $\left\lceil \frac{P_c + O_c}{M} \right\rceil$ is ignored to make Equation (4) differentiable. After obtaining a derivative formula of goodput with respect to P_c, we make the derivative value equal to zero. Thus, the corresponding value of P_c can be calculated, as shown in Equation (5):

$$P_{optimal} = \frac{(B * R + O_c)\ln(1 - r_p) + \sqrt{((B * R + O_c)\ln(1 - r_p))^2 - 4(B * R + O_c)M\ln(1 - r_p)}}{-2\ln(1 - r_p)} \tag{5}$$

where $P_{optimal}$ represents the optimal value of chunk size. This equation is valid when $r_p \in (0, 1)$.

3.2. **Segment transmission.** As shown in Figure 1(b), a chunk is divided into several segments. Each segment is suited to the MTU of links. A header is added to each segment. With segment headers, the transport layer of the destination can retain received segments. A segment is lost, which does not indicate that all segments will be discarded. The most typical example of segment transmission is ICTP [3]. The primary data transfer process can be summarized as follows. The receiver first sends a data request for the entire chunk, and the sender replies corresponding data. Then, the receiver resends requests for the entire chunk or the missing segments until the chunk is assembled.

Considering the overhead of each segment, the number of pieces that the chunk is split to can be written as follows:

$$N_p^{''} = \left\lceil \frac{P_c + O_c}{M - O_c} \right\rceil \tag{6}$$

where $N_p^{''}$ denotes the number of split pieces for segment transmission. In this case, packet loss does not fail the entire chunk, and retransmission is used to recover the lost segments quickly. Thus, we can acquire the entire chunk effectively. Supposing that the number of expected transmission for n segments is a_n, packet loss rate subject to $r_p \in (0, 1)$, then we get

$$\begin{cases} a_n = a_{n*r_p} + 1 \\ a_1 = \dfrac{1}{1 - r_p} \end{cases} \tag{7}$$

This result is derived in the form of a sequence but is essentially a continuous function of the a_n with respect to r_p and n. According to Equation (7), we have

$$a_n = \frac{1}{1 - r_p} + \log_{r_p} \frac{1}{n}$$

Because the number of segments that the chunk is split to is $N_p^{''}$, the number of expected transmission for the chunk $N_{ct}^{''}$ is

$$N_{ct}^{''} = \begin{cases} \dfrac{1}{1 - r_p} + \log_{r_p} \dfrac{1}{N_p^{''}}, & r_p > 0 \\ 1, & r_p = 0 \end{cases} \tag{8}$$

There are many retransmission policies for reliable transmission at the granularity of segments. In this paper, two of the most typical retransmission policies are chosen for analysis. Other retransmission policies can be approximately considered to be in between.

3.2.1. *Segment transmission with integral retransmission policy.* In this case, after receiving every segment, the receiver uses the segment data to assemble the chunk according to the offset information. The offset information is encapsulated in the segment header. Further, the receiver always resends the requests for the entire chunk until the chunk is assembled.

During the delivery of the chunk for the segment transmission mode with integral retransmission (SIM), the total bits needed to be sent are

$$Bits_{SIM} = N_{ct}^{''} * (P_c + O_c) \tag{9}$$

According to Equations (6), (8), and (9), the goodput of SIM is

$$Goodput_{SIM} = \frac{P_c}{Delay_{SIM}} = \frac{P_c}{N_{ct}^{''} * R + \frac{Bits_{SIM}}{B}} = \frac{P_c}{\left(R + \frac{P_c + O_c}{B}\right) N_{ct}^{''}} \tag{10}$$

where $N_{ct}^{''}$ are presented in Equation (8) and $Delay_{SIM}$ denotes the total time spent from sending the request to retrieving the entire chunk for SIM. Because the explicit expression is complex, the optimal chunk size will be obtained by using the numerical analysis method in the next section.

3.2.2. *Segment transmission with partial retransmission policy.* Although the integral retransmission policy is relatively simple and feasible, every retransmission of the entire chunk incurs a large bandwidth overhead. Especially with the development of hardware technology, the packet loss rate becomes smaller. Thus, the strategy of retransmitting partial segments is a better choice. In this case, the receiver resends the requests for the lost segments until the chunk is assembled, and the sender replies to the corresponding segment once receiving a request.

During the delivery of the chunk for the segment transmission mode with partial retransmission (SPM), the total bits needed to be sent are

$$Bits_{SPM} = N_p'' * M * \left(1 + r_p + \cdots + r_p^{N_{ct}''}\right) = N_p'' * M * \frac{1 - r_p^{N_{ct}''+1}}{1 - r_p} \tag{11}$$

The length of the unfilled parts of the last segment is much smaller than the chunk length. Thus, the deviation caused by unfilled parts is ignored in this paper. Then, the goodput of SPM is

$$\begin{aligned} Goodput_{SPM} &= \frac{P_c}{Delay_{SPM}} = \frac{P_c}{N_{ct}'' * R + \frac{Bits_{SPM}}{B}} \\ &= \frac{P_c}{N_{ct}'' * R + \frac{N_p'' * M * \left(1 - r_p^{N_{ct}''+1}\right)}{B * (1 - r_p)}} \end{aligned} \tag{12}$$

where N_p'' and N_{ct}'' are presented in Equations (6) and (8), respectively. Since Equation (12) is not differentiable, the optimal chunk size will be discussed in the next section.

In summary, we are the first to develop a mathematical model about three existing representative transmission modes. Different from the existing qualitative or experimental method, our quantitative method is based on mathematical theory. Moreover, the proposed model considers both the protocol overhead and network status in the process of content transfer and gives the optimal value of chunk size by maximizing goodput. It is noteworthy that our model is mainly built based on IP infrastructure but still works for future clean-slate approaches.

4. **Numerical Analysis.** Based on Equations (4), (10), and (12) in Section 3, we find that goodput is also related to the following six factors besides chunk size: chunk overhead, segment overhead, packet loss rate, MTU, RTT, and Bandwidth. However, for a given application and its running network, the MTU, chunk overhead and segment overhead are basically fixed. In this section, we mainly study the effects of chunk size on goodput under different RTTs, packet loss rates and bandwidths, and propose the recommended values of chunk size for some typical scenarios.

Under the current network environments, the max packet size is always bounded by Ethernet MTU, which is 1500 bytes by default, i.e., $M = 1500$ bytes. Taking the CCN as a reference, the overhead is in the order of 650 bytes per chunk [4], mostly due to security information (e.g., the signature needed to authenticate the chunk), i.e., $O_c = 650$ bytes. Moreover, the segment overhead is set to $O_s = 64$ bytes.

To study the effects of chunk size on the transfer performance in different network scenarios, we vary other factors based on the parameter space described in Table 2. We refer to [19,20] and set the network parameters as follows. Studies suggest that general servers experience a loss rate between 1% and 2% [21]. The packet loss rate has been greatly reduced with the development of network technology and infrastructure which could reach 0.5%, even 0.1% [20]. Thus, packet loss rate of 0.1%, 0.5%, 1% 2% and

TABLE 2. The parameter space in our analysis

Parameter	Description	Default value
Packet loss rate (r_p)	0, 0.1%, 0.5%, 1%, 2%	1%
RTT (R)	0ms, 1ms, 10ms, 20ms, 100ms	10ms
Bandwidth (B)	1Mbps, 10Mbps, 20Mbps, 50Mbps, 100Mbps	10Mbps

ideal packet loss rate zero are chosen. In the fourth generation (4G) cellular network, the typical RTT values are 100ms, and the 5G reduces it lower to 1ms [20]. However, end-to-end latencies in most scenarios are in between this range [22]. We choose 10ms and 20ms as the representative and the ideal RTT (0ms) is also considered. Finally, we choose bandwidths range from 1Mbps (3G) to 100Mbps (4G) [20], and several intermediate values such as 10Mbps, 20Mbps and 50Mbps are selected.

If there is no particular explanation, the values of other impact factors are the same as the default values except the variables to be studied in the following analysis.

4.1. **The performance of chunk transmission.** In this subsection, the performance of chunk transmission is evaluated under different packet loss rates, RTTs and bandwidths as shown in Figure 2. Moreover, the values of the optimal chunk size point are marked except when the packet loss rate is zero.

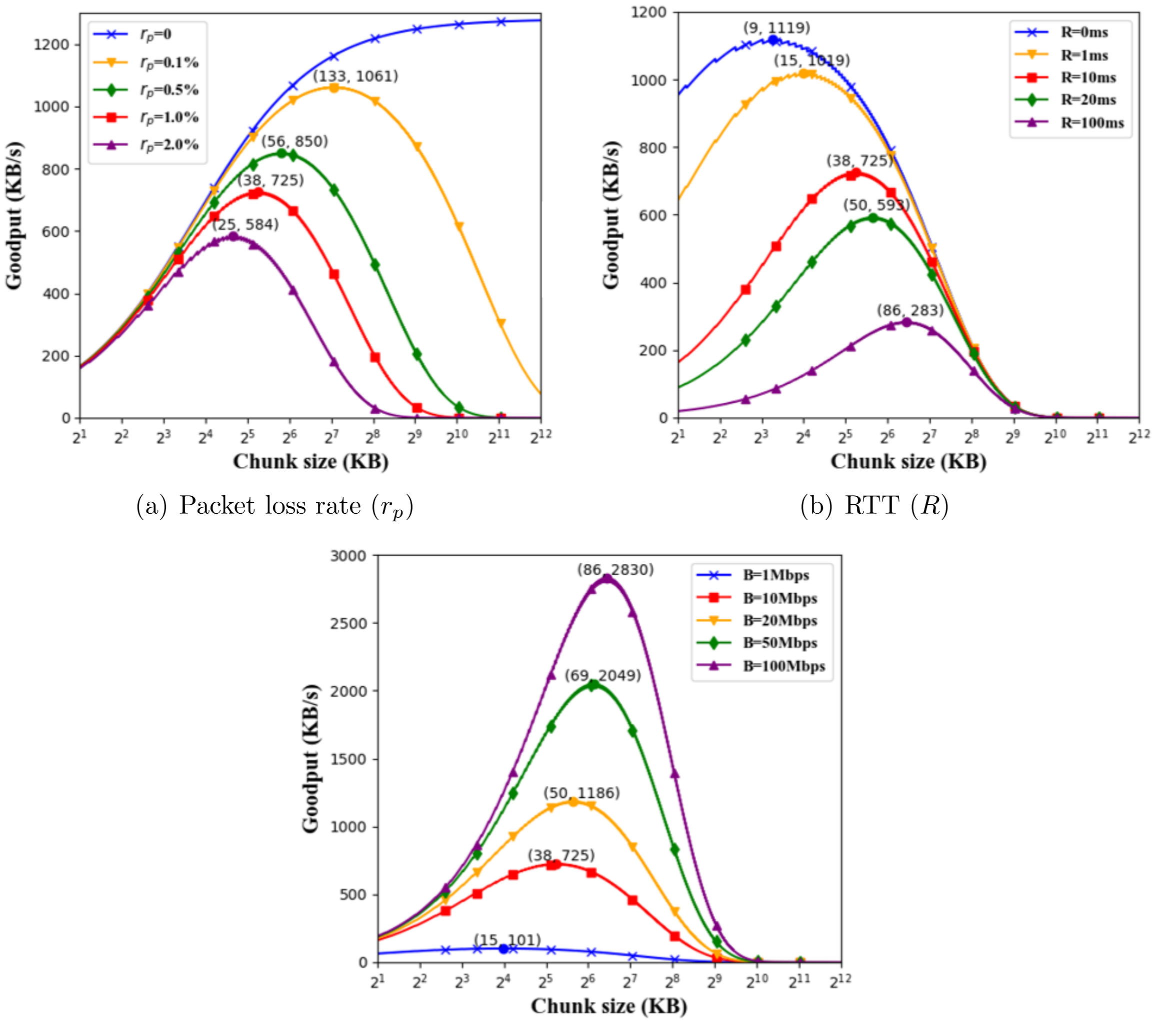
We observe that the performance of CTM all deteriorates and always goes down to zero when the chunk size exceeds a certain threshold as long as the packet loss rate is not zero. This is because packet loss leads to the retransmission of the entire chunk. In addition, three factors all have a significant effect on goodput, and the maximum goodput can reach 80% of the bandwidth only when the network is in good condition. In our parameter space, the recommended values of the chunk size can be dozens of KBs (e.g., 16, 32, and 64) up to 128KB.

With the effect of packet loss, the performance of CTM deteriorates very seriously as shown in Figure 2(a). When the chunk size is small, the chunk overhead is the main factor that reduces the transmission efficiency since the 650 bytes is a large overhead ratio. Intuitively, the increase of packet loss rate causes a decrease in the optimal chunk size, that is to say, the change of peak point. This figure also shows that the optimal chunk size is 38KB when the packet loss rate is 1%. If the value of size is 2^n, where n is an integer, then the value of optimal chunk size is recommended as 32KB.

Considering Figure 2(b), we can see that the chunk size can be bigger for better performance with the increase of RTT. This is because transmitting more data once can reduce the end-to-end delay overhead ratio. Besides, the higher the RTT, the higher the transmission overhead, and hence the lower the goodput.

Figure 2(c) illustrates that the larger the bandwidth, the bigger the chunk size that reaches the maximum goodput. Bandwidth effectively increases the value of the optimal chunk size. The reason is that the larger the bandwidth, the larger the amount of data that is allowed to be carried while other given factors are fixed. Moreover, with the increase of bandwidths, the proportion of the maximum goodput in the bandwidth decreases. It shows that CTM has low bandwidth utilization.

Figure 2 shows the numerical results are consistent with the intuitive understanding. For example, packet loss leads to the retransmission of the entire chunk for CTM. Thus, the transfer performance becomes lower with the increasing of chunk size when the chunk size exceeds a certain threshold. The numerical results show our model can describe variation trends of the optimal chunk size in CTM.



(a) Packet loss rate (r_p)

(b) RTT (R)

(c) Bandwidth (B)

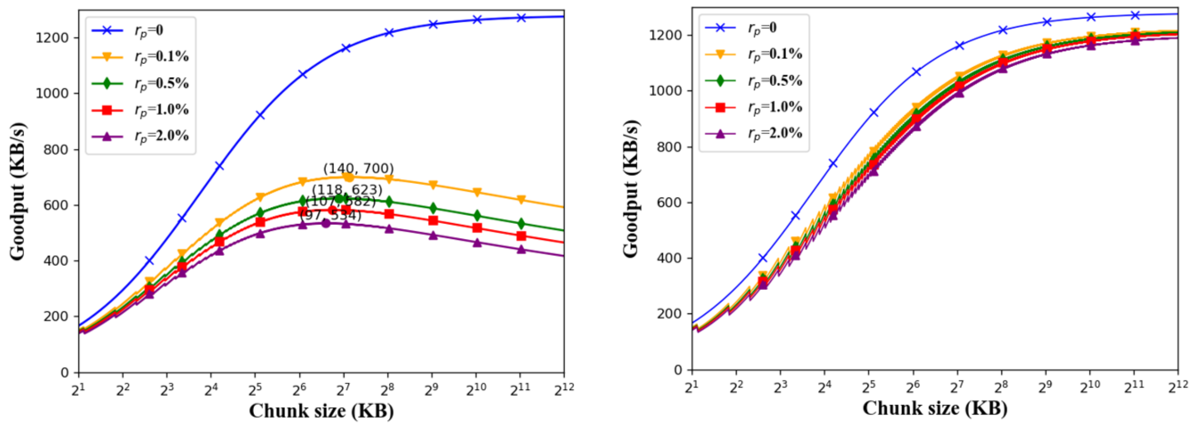
FIGURE 2. Performance trends of CTM for three factors

4.2. **The performance of segment transmission.** Under different packet loss rates, RTTs, and bandwidths, we consider the effects of chunk size on segment transmission with two different retransmission policies.

As we can see from Figure 3, the packet loss rate has little effect on the optimal chunk size for segment transmission. When the chunk size comes to a specific value such as 64KB, the goodput of SIM decays slowly, and SPM can still keep a considerable goodput. Compared to chunk transmission, the segment transmission can effectively reduce the effect of packet loss rate on goodput and support bigger chunk size.

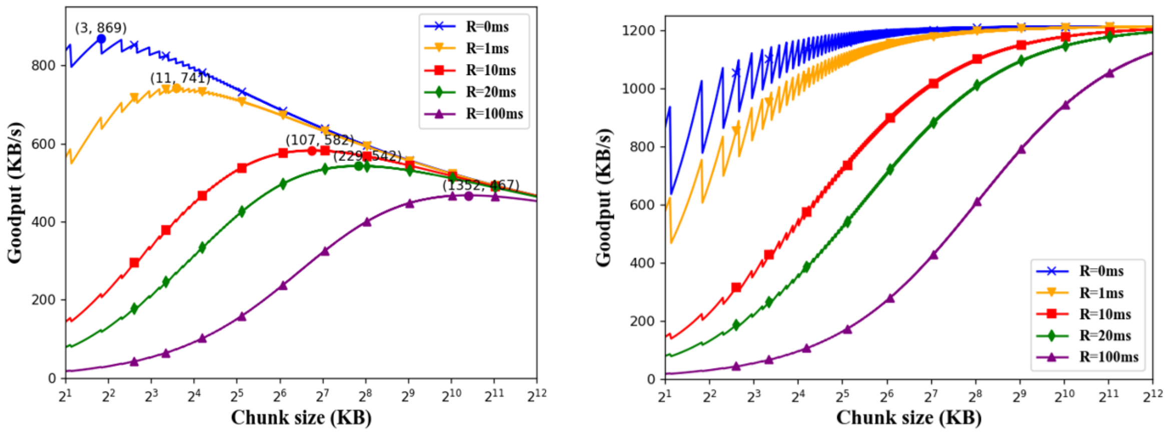
The packet loss rate has a very limited effect on SPM as shown in Figure 3(b). After the chunk size exceeds a certain threshold, goodput gradually approaches the upper value. Using the complex response logic and computational overhead of the sender, SPM can reduce the bandwidth costs to achieve better performance.

Figure 4 shows the effect of RTT on goodput gradually tends to be consistent when the chunk size is large, with no significant difference in affecting the transmission efficiency. The reason is that the RTT has less effect on goodput when the chunk size is large. When RTT increases, the optimal chunk size selected in Figure 4(a) is also relatively larger, and a larger chunk size is also needed in Figure 4(b) to achieve better performance.



(a) SIM (b) SPM

FIGURE 3. Effect of packet loss rate (r_p) on SIM and SPM



(a) SIM (b) SPM

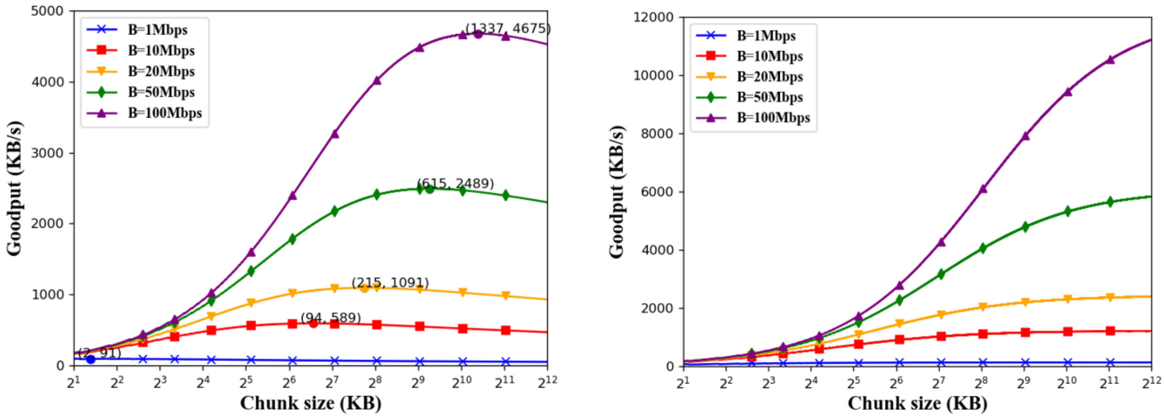
FIGURE 4. Effect of RTT (R) on SIM and SPM

Figure 5 illustrates that the increase in bandwidth can significantly improve goodput performance. The bandwidth also greatly affects the value of the optimal chunk size. As shown in Figure 5(b), the bandwidth cost is relatively small because only parts of the segments are retransmitted. Thus, goodput increases exponentially with the chunk size.

In this sub-section, Figures 3, 4, and 5 show the effects of the packet loss rates, RTTs, and bandwidths on segment transmission, respectively. We find that the optimal chunk size can range from 4KB to 1MB for segment transmission. Besides, considerable goodput is maintained, even if a large chunk size is chosen. The results illustrate that the trends of the optimal chunk sizes are well-founded under different network circumstances.

4.3. **Comparison and analysis.** As shown in Figure 6, we compare the proposed three transmission modes with the default parameter setting and ideal transmission (chunk transmission with both packet loss rate and RTT equal to zero).

The results show that the optimal chunk size and the goodput of CTM are always relatively small as long as the packet loss rate is not zero. This is because packet loss leads to the retransmission of the entire chunk. In the scenario with the default setting, the recommended chunk size is about 32KB, and the maximum goodput of CTM only



(a) SIM (b) SPM

FIGURE 5. Effect of bandwidth (B) on SIM and SPM

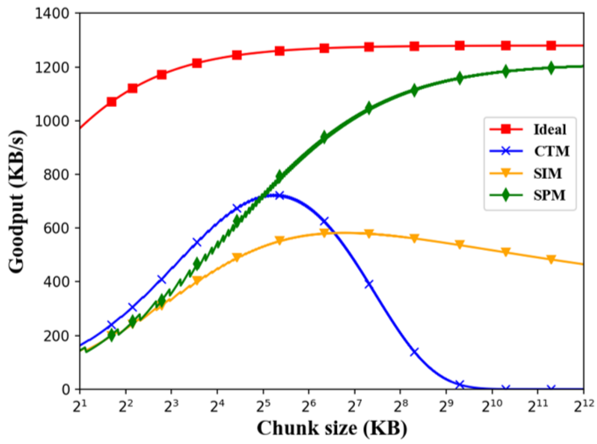


FIGURE 6. Comparison of the proposed three transmission modes and ideal transmission. Both the packet loss rate and RTT of ideal transmission are equal to zero. Other parameters are all default values.

reaches about sixty percent of the bandwidth. In contrast, the segment transmission can effectively reduce the effect of packet loss rate on goodput. The segment transmission outperforms chunk transmission in most cases, and the goodput of segment transmission is not going down rapidly when the chunk size is big enough.

The goodput performance of SPM is much closer to ideal transmission than SIM, especially when the chunk size is big. The reason is that there are more redundant segments when using the integral retransmission policy. The more data are transmitted, the lower the goodput is. Besides, the protocol overhead is the main factor affecting the transmission efficiency of a small chunk size for SPM.

For segment transmission, the transmission efficiency is still very high even if the chunk size is big. And the change of chunk size has less effect on goodput for segment transmission. This feature leaves more space for a tradeoff in other factors (e.g., caching benefits). However, plenty of resources, such as bandwidth (caused by protocol overhead) and computation (caused by timer and verification, etc.), are used to maintain the transport service of ICN. The costs of these resources are not negligible.

Based on the above analysis, we can draw the following conclusions. When an ICN architecture is compatible with IP basic infrastructure, we recommend to use the segment transmission for better transmission efficiency, and the chunk size can be set to greater than 1MB. Besides, with the development of the clean-slate ICN infrastructure, the MTU becomes larger, and the packet loss rate gets lower. Thus, chunk transmission also can achieve excellent performance, and the chunk size can be set to 128KB.

In this section, we conducted numerical simulations about three transmission modes with a wide range of network parameters that can override most current classic network scenarios. We analyzed the trends of the optimal chunk size and gave the corresponding explanations. The results illustrate that our model is meaningful for maximizing transmission efficiency by providing an optimal chunk size.

5. **Experiment Results and Discussion.** In the last section, the proposed model has been studied by numerical analysis. We plotted figures about the effect of different factors on the optimal value of chunk size and drew several conclusions. However, our analysis is based on idealized network conditions. The correctness of the model has not been fully validated. We still need to conduct real-world transmission experiments to find out aspects that we do not take into account. In this section, we present the experimental studies to verify the accuracy of our model further and make a discussion about the gap between experimental results and that derived from our numerical analysis.

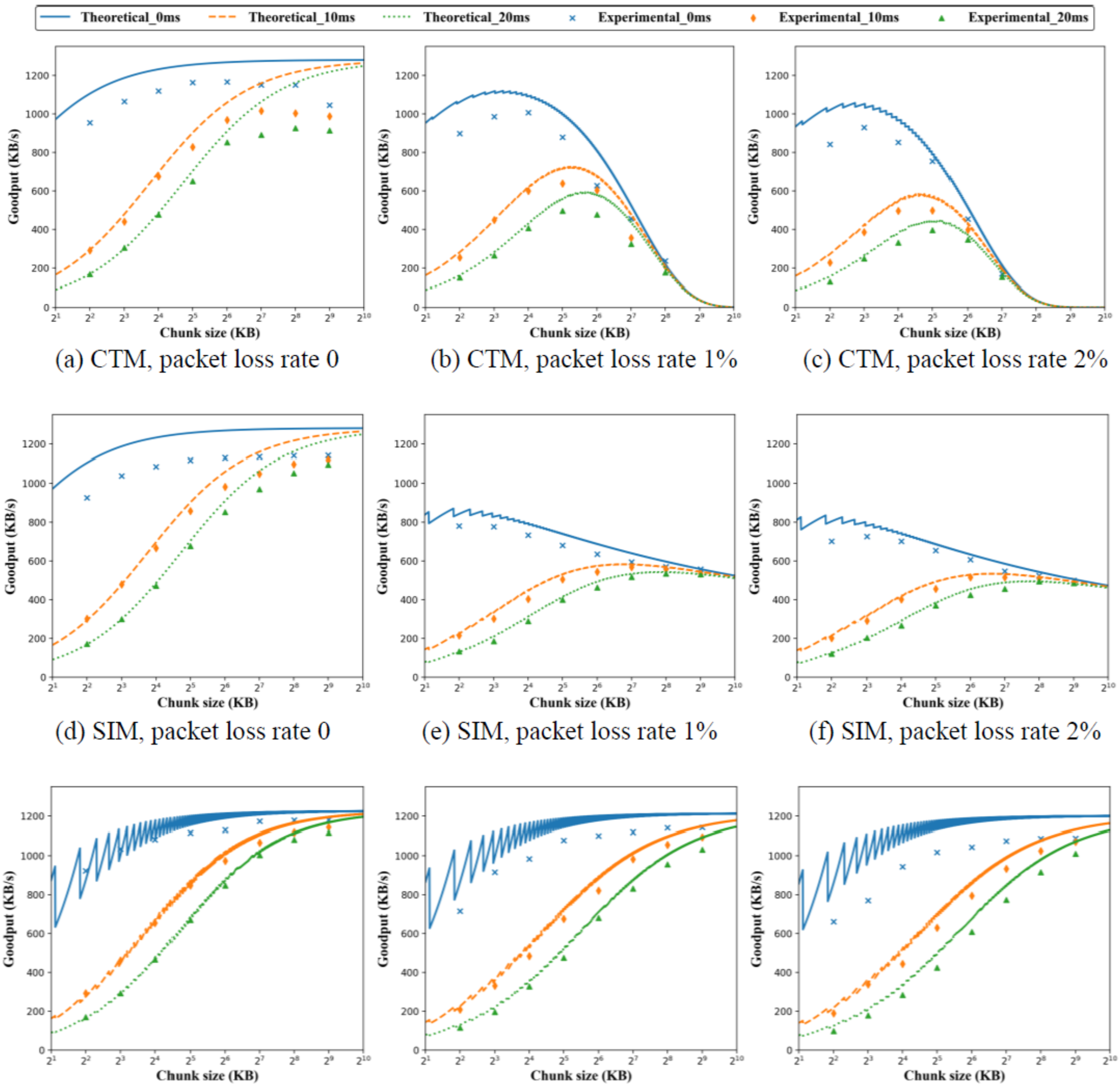
The experiments are conducted by setting up a receiver and a sender. The operating system on these two devices is Ubuntu16.04. The sender and receiver link capacities are 100Mb/s with a very small (μs) delay and almost no loss. We set $O_c = 650$ bytes, $O_s = 64$ bytes and $M = 1500$ bytes, as described in Section 4. The encapsulation is measured: into 1500 byte IP packets (with 20 bytes IPv4 header and 64 bytes segment header) using a payload size of 1416 bytes.

In our experiments, the emulation tool NetEm [23] is used to vary network parameters and limit bandwidth. The bandwidth is limited to 10Mbps to avoid large errors or import background traffic. The sender sends the entire chunk by means of irrigation to avoid introducing a special congestion control algorithm and maximize the utilization of bandwidth. For simplicity, only packet loss rates in {0, 1%, 2%} and RTTs in {0ms, 10ms, 20ms} are picked as representative parameters to verify the model. We measure the time needed to transfer a 1MB file via three transmission modes under different packet loss rates and RTTs. We repeat our experiments ten times and present the average values of goodput to exclude the effect of random loss. The results are shown in Figure 7.

As we can see from Figures 7(a), 7(b), and 7(c), because the packet loss rate of the experimental physical link is not zero, this severely reduces the experimental goodput of CTM. Similarly, it also has a great effect on three transmission modes, especially if the packet loss rate is 0, as shown in Figures 7(a), 7(d), and 7(g). Besides, the errors between the experimental and theoretical results are small when the packet loss rate equals 1% or 2%, as shown in Figures 7(e) and 7(f).

Figures 7(g), 7(h), and 7(i) show the experimental errors due to the traversal detection of the lost segments in SPM. The larger the chunk size is, the more retransmission is, the higher the traversal detection cost is, and thus, the greater the experimental and theoretical differences are. When the chunk size is small, the IP protocol overhead accounts for a larger proportion of the chunk size. Thus, the bigger experimental errors are. Moreover, for the RTT impact factor, the experimental errors between 10ms and 20ms are relatively smaller than that between 0ms and 10ms.

It is obvious that the measured goodput is slightly lower than the predicted value of the model. It can be caused by the computation overhead, the IP protocol overhead, the initial RTT and packet loss rate of a link, the unfilled parts of segments, and so on. Taking these impact factors into account, the errors between the real-world experimental results



(a) CTM, packet loss rate 0 (b) CTM, packet loss rate 1% (c) CTM, packet loss rate 2%

(d) SIM, packet loss rate 0 (e) SIM, packet loss rate 1% (f) SIM, packet loss rate 2%

(g) SPM, packet loss rate 0 (h) SPM, packet loss rate 1% (i) SPM, packet loss rate 2%

FIGURE 7. Experimental and theoretical results under different packet loss rates and RTTs. The packet loss rates are 0, 1%, and 2%. The RTTs are 0ms, 10ms, and 20ms.

and that of numerical are small for the different network parameters we set. Overall, the accuracy of the proposed model is verified by the experiments. It further illustrates that our model can provide a meaningful reference to the optimal chunk size for multiple network scenarios.

6. **Conclusions.** In this paper, a mathematical model was developed by considering both the protocol overhead and network status. Three typical transmission modes are analyzed, and the optimal chunk size was derived by maximizing the goodput of content transfer in ICN. Both experimental and numerical methods verified the accuracy of the proposed model. The recommended values of the chunk size are also provided for two deployment approaches in ICN. Moreover, our model is also applicable to future clean-slate approaches with different network status.

In ICN, in-network caching is an important topic to be investigated. Small chunk size is great for caching benefits to avoid the inaccuracy between content popularity and cached contents. Hence, there is a tradeoff between caching benefits and transmission efficiency. Therefore, our future work will focus on proposing an algorithm to search for the suitable values of the chunk size.

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